

Volume I · Chapter 5 — Medical NLP Basics (regex → embeddings) · Labs

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-I-Chapter-5-context.md` for the AI interaction log.

Prerequisites: Ch 1 (Python/Pandas).

Learning outcomes — the student can:

- Process clinical free text with **classical NLP** (tokenization, normalization, TF-IDF).
- Build **regex/rule-based extractors** for clinical concepts and detect **negation**.
- Use **word embeddings** (general and biomedical) and build a simple clinical **text classifier**.
- Articulate the **limits of classical NLP** (context, negation) that motivate transformers (bridge to Volume II).

Labs (hands-on) — 10 h:

#	Lab	h
5a	Preprocess clinical notes: tokenize, normalize, build a clean corpus	2
5b	Regex extractor: pull vitals / medications / dates from clinical text	3
5c	TF-IDF + logistic regression: classify clinical note type/intent	3
5d	Explore word embeddings: nearest neighbors on biomedical terms	2

Datasets/tools: synthetic/de-identified clinical notes; Python `re`, `scikit-learn`, `spaCy`/`NLTK`, `gensim` or pretrained biomedical embeddings.

Assessment: regex extractor deliverable — rubric (**40%**); text-classification lab (**40%**); quiz (**20%**).

Key decisions: rule-based vs. statistical NLP; when regex suffices vs. needs ML; **general vs. biomedical** embeddings; handling **negation/context**.

References: plan §13 → *Transformers & the Hugging Face ecosystem* (bridge); *Programming & data science*.

Hours: Theory **10** + Lab **10** = **20**.